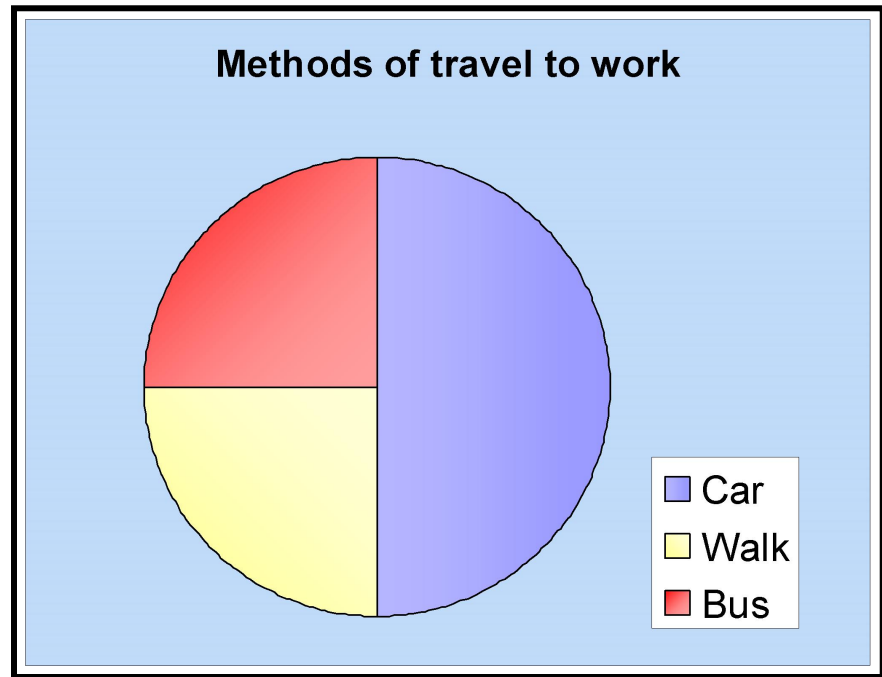


Pie charts

A **pie chart** is a circle divided up into sectors which are representative of the data.

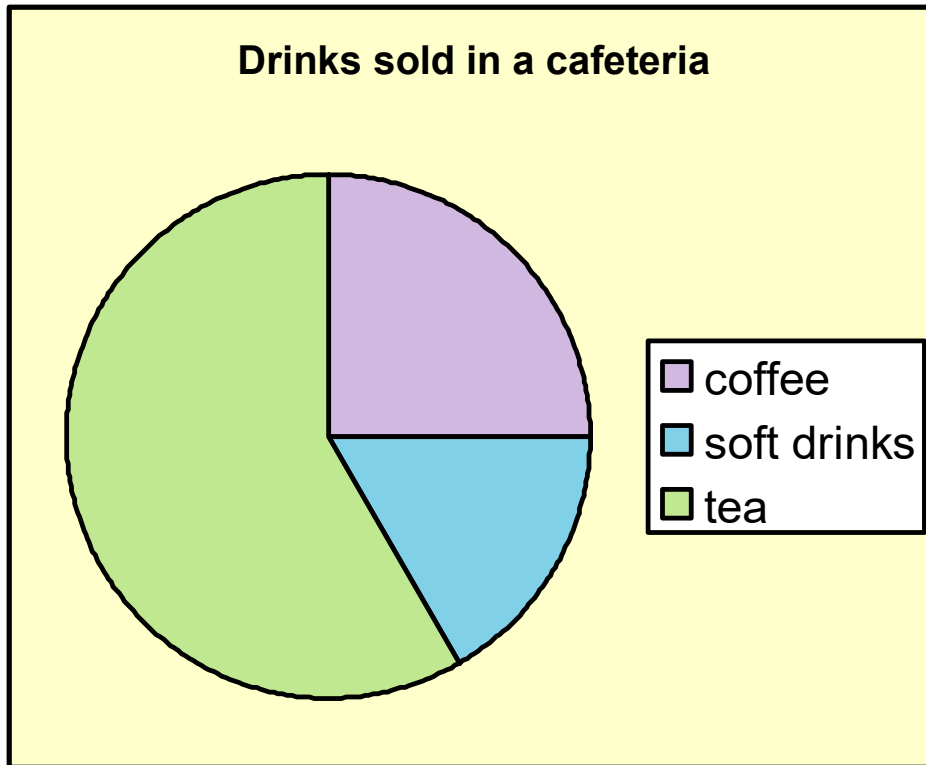
In a pie chart, each category is shown as a fraction of the circle.

For example, in a survey half the people asked drove to work, a quarter walked and a quarter went by bus.



Pie charts

This pie chart shows the distribution of drinks sold in a cafeteria on a particular day.



Altogether 300 drinks were sold.

Estimate the number of each type of drink sold.

Coffee: 75

Soft drinks: 50

Tea: 175

Drawing pie charts

To draw a pie chart you need compasses and a protractor.

The first step is to work out the angle needed to represent each category in the pie chart.

We need to work out how many degrees are needed to represent each person or thing in the sample.

Drawing pie charts

For example, 30 people were asked which newspapers they read regularly.

The results were :

Newspaper	Number of people
<i>The Guardian</i>	8
<i>Daily Mirror</i>	7
<i>The Times</i>	3
<i>The Sun</i>	6
<i>Daily Express</i>	6

Drawing pie charts

Method 1

There are 30 people in the survey and 360° in a full pie chart.

Each person is therefore represented by $360^\circ \div 30 = 12^\circ$

We can now calculate the angle for each category:

Newspaper	No of people	Working	Angle
<i>The Guardian</i>	8	$8 \times 12^\circ$	96°
<i>Daily Mirror</i>	7	$7 \times 12^\circ$	84°
<i>The Times</i>	3	$3 \times 12^\circ$	36°
<i>The Sun</i>	6	$6 \times 12^\circ$	72°
<i>Daily Express</i>	6	$6 \times 12^\circ$	72°
Total	30		360°

Drawing pie charts

Once the angles have been calculated you can draw the pie chart.

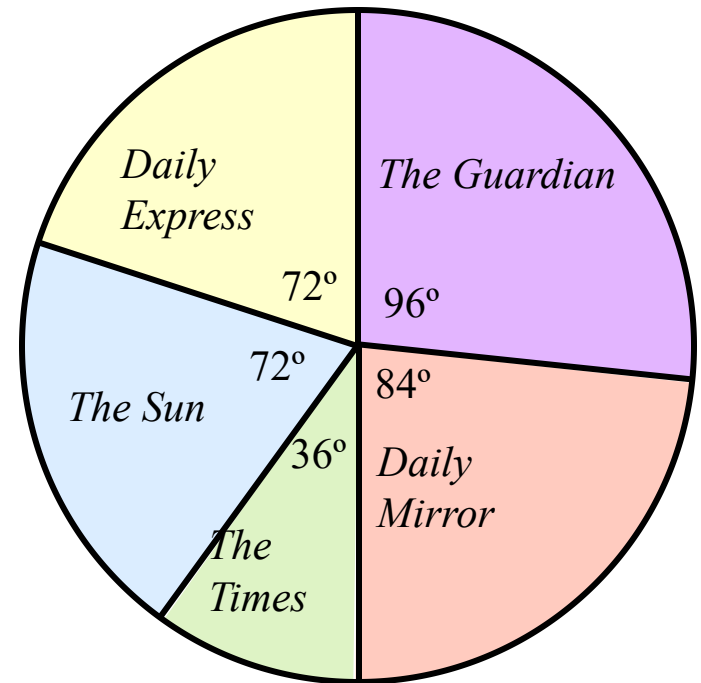
Start by drawing a circle using compasses.

Draw a radius.

Measure an angle of 96° from the radius using a protractor and label the sector.

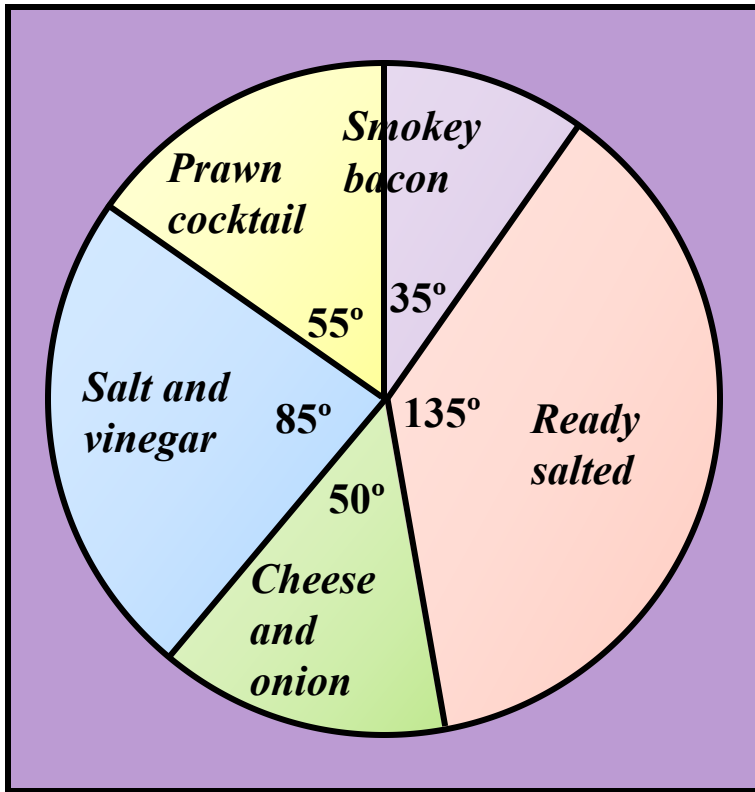
Measure an angle of 84° from the last line you drew and label the sector.

Repeat for each sector until the pie chart is complete.



Reading pie charts

The following pie chart shows the favourite crisp flavours of 72 children.



How many children preferred ready salted crisps?

How many degrees represents one child?

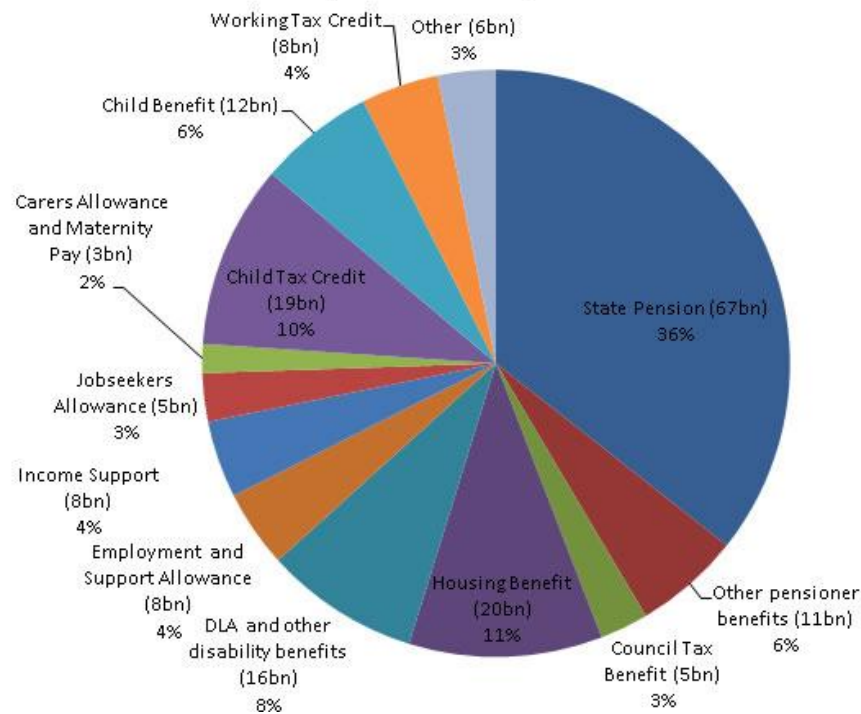
$$\frac{360}{72} = 5^\circ.$$

The number of children who preferred ready salted is:

$$135 \div 5 = 27$$

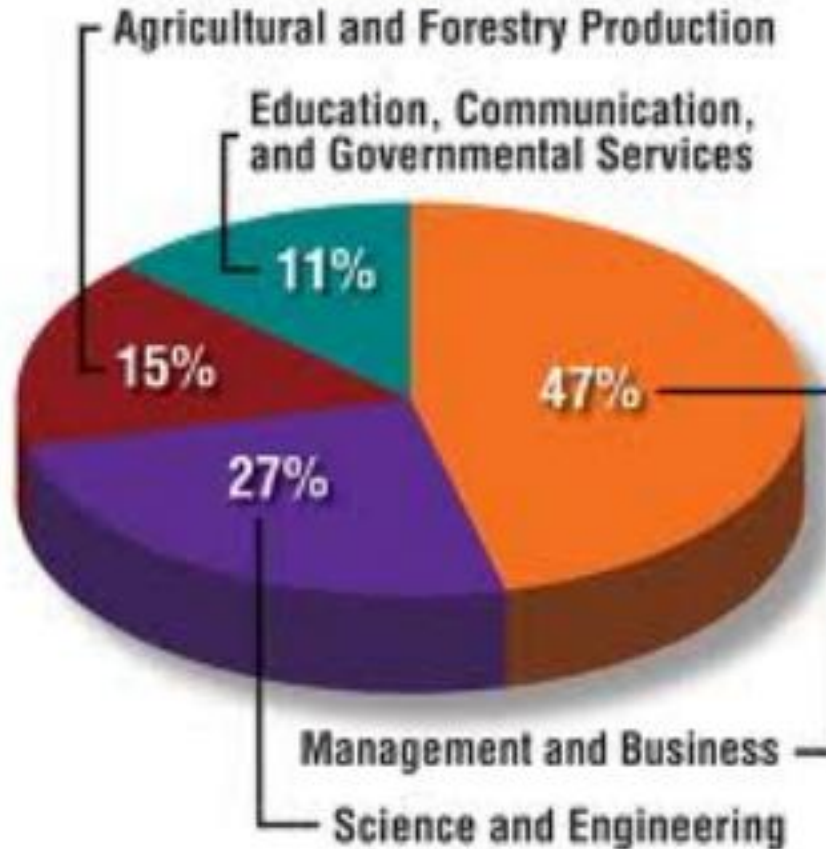
Pie Chart

Breakdown of Welfare Spending, 2009–10
(Total 188bn)

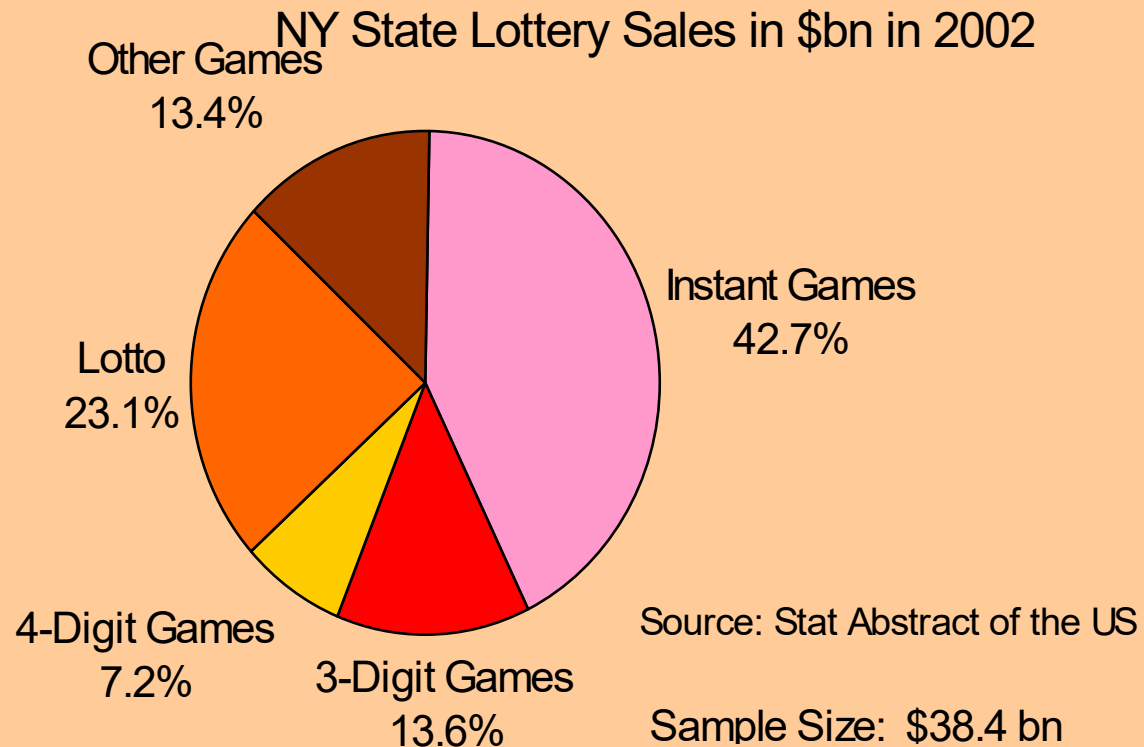


Pie Chart

Employment Opportunities



Pie to Bar: Spending On NY Lottery



Spending On NY Lottery

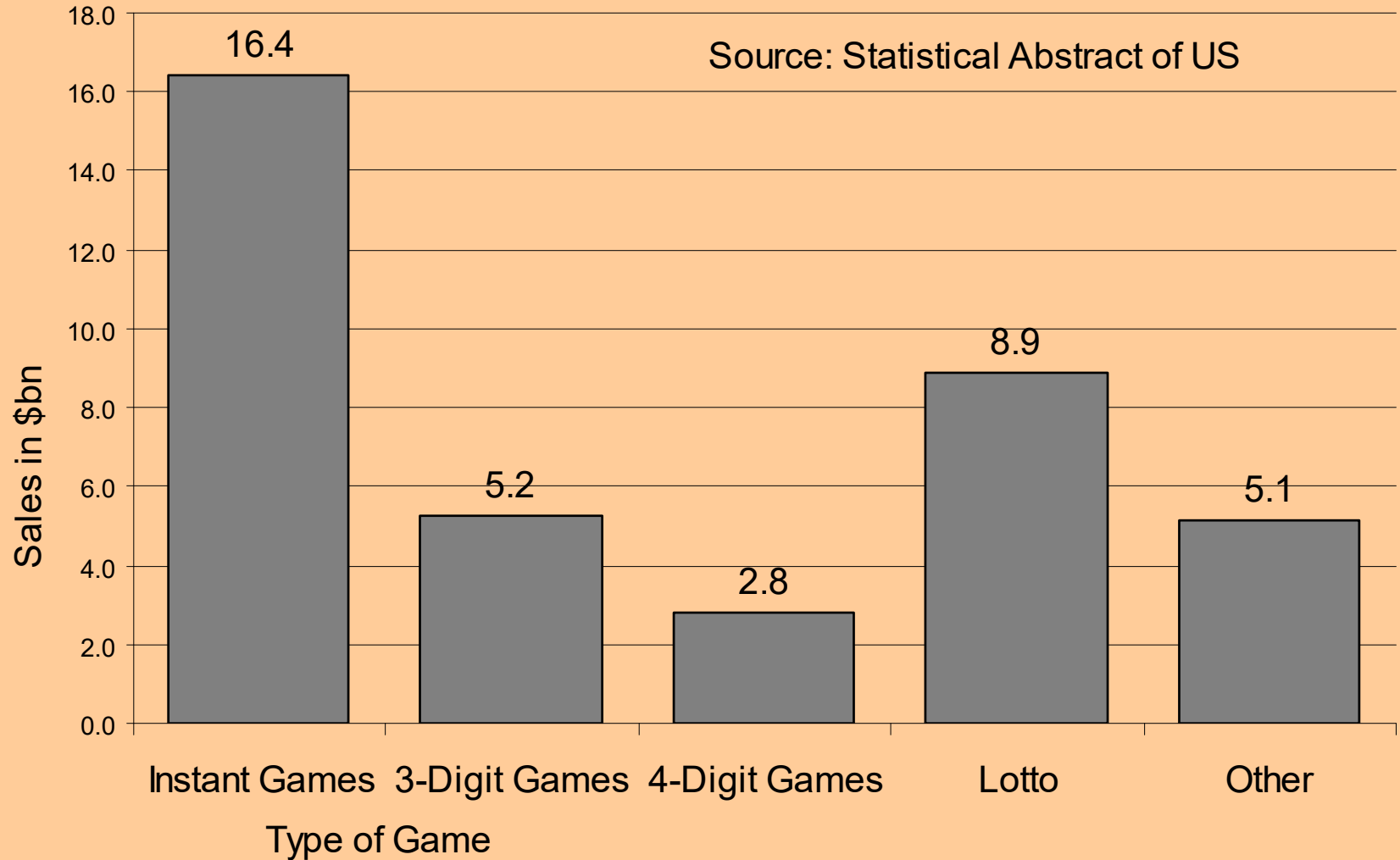
Type of Game	\$ in bn	% nrst 0.1	∠ ° nrst 0.1
Instant Games			°
3-Digit Games			°
4-Digit Games			°
Lotto			°
Other Games			°
Totals			°

Spending On NY Lottery

Type of Game	\$ in bn	% nrst 0.1	∠ ° nrst 0.1
Instant Games	16.4	42.7%	153.8 °
3-Digit Games	5.2	13.6%	49.1 °
4-Digit Games	2.8	7.2%	26.0 °
Lotto	8.9	23.1%	83.0 °
Other Games	5.1	13.4%	48.1 °
Totals	38.4	100.0%	360.0 °

NY State Lottery Sales in \$bn

Source: Statistical Abstract of US



Turning a Pie Chart into a Bar Graph

